

SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA14

COURSE NAME: Ethical Hacking

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COURSE OUTCOME COVERED

CO3: Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)

QUESTIONS: 5

1. A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server **scanme.nmap.org**. Before conducting any vulnerability assessment, the client requires a detailed enumeration report. Using **Nmap**, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on **scanme.nmap.org**. Analyze all identified ports and services, determine their business purpose, and evaluate whether they are necessary for public access. Discuss how a threat actor could utilize the information gathered during enumeration to launch further attacks such as service exploitation, brute-force attacks, or vulnerability discovery. Prepare a technical report summarizing your findings, attack surface analysis, risk assessment, and recommendations for securing the server.
2. The security team has deployed a deliberately vulnerable Linux server, **192.168.56.101 (Metasploitable 2)**, in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use **Nmap** to perform comprehensive enumeration of **192.168.56.101**. Identify all open ports, running services, service versions, and operating system information. Analyze the purpose of each service and determine which services could provide potential entry points for an attacker. Categorize the discovered services into High, Medium, and Low risk levels based on their exposure and functionality. Explain how the enumeration results could guide a penetration tester in selecting potential attack vectors and exploitation strategies.
3. A financial organization wants to assess the security posture of a Windows workstation located at **192.168.56.102** before connecting it to the production network. Using **Nmap**, enumerate **192.168.56.102** to identify active ports, network services, file-sharing protocols, remote management services, and operating system details. Analyze whether the discovered services comply with the principle of least privilege and identify any services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement, or unauthorized access. Based on your findings, propose a set of security hardening measures to improve the system's resilience against cyberattacks.

4. An organization hosts a web application on **192.168.56.103 (OWASP Juice Shop Server)** and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using **Nmap**, perform a detailed enumeration of **192.168.56.103**, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in the web application environment and recommend measures to minimize exposure
5. The network gateway device at **192.168.56.1** serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using **Nmap**, conduct a comprehensive enumeration of **192.168.56.1**, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.

QUESTION1:

1. Host Discovery

`nmap -sn scanme.nmap.org`

2.

```
Session Actions Edit View Help
pujitha@POOJITHA: ~
{pujitha@POOJITHA:~
$ nmap -sn scanme.nmap.org

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:02 IST
Nmap scan report for scanme.nmap.org (scanme.nmap.org)
Host is up (0.028s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.41 seconds
```

TCP Port Scan

`nmap -sS -p- scanme.nmap.org`

```
Session Actions Edit View Help
pujitha@POOJITHA: ~
{pujitha@POOJITHA:~
$ nmap -sS -p- scanme.nmap.org

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:05 IST
Nmap scan report for scanme.nmap.org (scanme.nmap.org)
Host is up (0.031s latency).
Not shown: 65531 closed tcp ports (reset)

PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp   open  nping-echo
31337/tcp  open  tcpwrapped

Nmap done: 1 IP address (1 host up) scanned in 14.87 seconds
```

3. Service Version Detection

`nmap -sV -p22,80,9929,31337 scanme.nmap.org`

```
Session Actions Edit View Help
pujitha@POOJITHA: ~
{pujitha@POOJITHA:~
$ nmap -sV -p22,80,9929,31337 scanme.nmap.org

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:09 IST
Nmap scan report for scanme.nmap.org (scanme.nmap.org)
Host is up (0.027s latency).

PORT      STATE SERVICE          VERSION
22/tcp    open  ssh              OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http             Apache httpd 2.4.7 ((Ubuntu))
9929/tcp   open  nping-echo       Nping echo
31337/tcp  open  tcpwrapped

Service detection performed. Please report any incorrect results...
Nmap done: 1 IP address (1 host up) scanned in 11.53 seconds
```

QUESTION 2:

1. Host Discovery

`nmap -sn 192.168.56.101`

```
pujitha@POOJITHA: ~  
Session Actions Edit View Help  
[pujitha@POOJITHA:~  
$ nmap -SS -p- scanme.nmap.org  
  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:05 IST  
Nmap scan report for scanme.nmap.org (scanme.nmap.org)  
Host is up (0.031s latency).  
Not shown: 65531 closed tcp ports (reset)  
  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
9929/tcp  open  nping-echo  
31337/tcp open  tcpwrapped  
  
Nmap done: 1 IP address (1 host up) scanned in 14.87 seconds
```

2. TCP Port Scan

`nmap -sS -p- 192.168.56.101`

```
pujitha@POOJITHA: ~  
Session Actions Edit View Help  
[pujitha@POOJITHA:~  
$ nmap -sn scanme.nmap.org  
  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:02 IST  
Nmap scan report for scanme.nmap.org (scanme.nmap.org)  
Host is up (0.028s latency).  
Nmap done: 1 IP address (1 host up) scanned in 2.41 seconds
```

QUESTION 3:

1. Host Discovery

`nmap -sn 192.168.56.102`

```

[pujitha@POOJITHA: ~]
$ nmap -sn 192.168.56.102

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 14:41 IST
Nmap scan report for 192.168.56.102
Host is up (0.00031s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.21 seconds

```

2.TCP Port Scan

`nmap -sS -p- 192.168.56.102`

```

[pujitha@POOJITHA: ~]
$ nmap -sS -p- 192.168.56.102

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 14:42 IST
Nmap scan report for 192.168.56.102
Host is up (0.00018s latency).
Not shown: 65527 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3306/tcp  open  mysql
5357/tcp  open  wsddapi

Nmap done: 1 IP address (1 host up) scanned in 24.87 seconds

```

QUESTION 4:

1. Host Discovery

`nmap -sn 192.168.56.103`

```

[pujitha@POOJITHA: ~]
$ nmap -sn 192.168.56.103

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 14:41 IST
Nmap scan report for 192.168.56.102
Host is up (0.00031s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.21 seconds

```

2. Service Version Detection

`nmap -sV -p22,3000 192.168.56.103`

```
Session Actions Edit View Help
(pujitha@POOJITHA)~
$ nmap -sV -p- 192.168.56.102

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 14:42 IST
Nmap scan report for 192.168.56.102
Host is up (0.00018s latency).
Not shown: 65527 closed tcp ports (reset)

PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3389/tcp  open  ms-wbt-server
5357/tcp  open  http

Nmap done: 1 IP address (1 host up) scanned in 14.87 seconds
```

QUESTION 5:

1. Host Discovery

`nmap -sn 192.168.56.1`

```
Session Actions Edit View Help
(pujitha@POOJITHA)~
$ nmap -sV -p135,139,445,3389,5357 192.168.56.102

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 14:43 IST
Nmap scan report for 192.168.56.102
Host is up (0.00022s latency).

PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds  Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.62 seconds
```

2. Script-Based Enumeration

`nmap -sC -p22,23,53,80,443 192.168.56.1`

```
Session Actions Edit View Help
(pujitha@POOJITHA)~
$ nmap -sC -p135,139,445,3389,5357 192.168.56.102

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 14:45 IST
Nmap scan report for 192.168.56.102
Host is up (0.00026s latency).

PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds  Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 25.16 seconds
```


